

RESEARCH INTERESTS

- Distributional data analysis, optimal transport (primary application: wearable device data)
- High-dimensional compositional data analysis (primary application: microbiome data)
- Non-Euclidean data analysis, sufficient dimension reduction, kernel methods

PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow

09/2024 –

Department of Biostatistics, University of Michigan, Ann Arbor, USA
Advisor: Irina Gaynanova

EDUCATION

Ph.D. in Mathematical Sciences

2024

KAIST, Daejeon, Korea

Thesis: “Kernel Methods for Compositional Data and Dimensionality Reduction”

Advisors: Cheolwoo Park, Jeongyoun Ahn

- Focus: Statistics (08/2021 – 02/2024), Algebraic Geometry (03/2018 – 07/2021)
- Continued postdoctoral research in Statistics with the same advisors for 5 months (03 – 08/2024)

B.S. in Mathematics

2018

Korea University, Seoul, Korea

AWARDS AND HONORS

Travel Awards/Invitations

- **IMS New Researchers Conference**, UMass Amherst, MA, USA 2026
- **IMS New Researchers Travel Award** 2026
- **Long Program on Connectomics**, Institute of Mathematical and Statistical Innovation (IMSI) 2026
- **1st IMS International New Researchers Conference**, University of Seville, Spain 2025

Awards and Scholarships in Korea

- **Best Ph.D. Dissertation Award**, Korean Statistical Society (prize: 3,000,000 KRW) 2025
- **Presentation Award for Students**, 2nd place, Korean Statistical Society (prize: 300,000 KRW) 2022
- **Outstanding Teaching Assistant Award**, Calculus II, KAIST Fall, 2020
- **University Students Contest of Mathematics**, Silver Award, Korean Mathematical Society 2016, 2017
- **Presidential Science Undergraduate Fellowship**, fully funded for 8 semesters 2013–2017
- **The Korean Mathematical Olympiad (KMO)** Gold Award, 2nd round 2009

RESEARCH PAPERS

* denotes the co-first authorship. Underline denotes mentored Master/Ph.D. student

Preprints/Submitted:

9. **Park, J.** and Gaynanova, I. (2026+) “Fréchet regression of multivariate distributions with nonparanormal transport.” Preprint available on [arXiv:2603.07014](https://arxiv.org/abs/2603.07014). R package [NonparaTransport](#).
- 2026 IMS New Researchers Travel Award
8. Shao, E.*, **Park, J.***, Punjabi, N., Jiang, H., and Gaynanova, I. (2026+) “Fast distance computation between multivariate distributions via nonparanormal Transport.” Under ‘Revise and Resubmit’ in the *Journal of Computational and Graphical Statistics* (planned resubmission: Oct. 2026). Preprint available on [arXiv:2603.00322](https://arxiv.org/abs/2603.00322). R package [NonparaTransport](#). (Shao E. received 2026 IMS Hannan Graduate Student Travel Award)
7. **Park, J.**, Park, C., and Ahn, J. (2026+) “Geometry-preserving and interpretable dimension reduction of compositional data.” Preprint available on [arXiv:2509.05563](https://arxiv.org/abs/2509.05563).

Peer-reviewed publications:

6. Xu, C., Schmidt, B.M., Krambrink, A., **Park, J.**, Song, P.X.-K., Jones, T.L.Z., Holmes, C., Chen, K., Ye, W., Kolenic, G., Roy, S., Sen, C., Pop-Busui, R., Spino, C., and Gaynanova, I. (2026) “Longitudinal Healing and Amputation Trajectories in Diabetic Foot Ulcers: Predictive Power of Wound Area and Duration and Sample-Size Implications From the Diabetic Foot Consortium.” *Wound repair and regeneration*, 34, No. 4, e70189. [Link](#).
5. **Park, J.**, Kok, N., and Gaynanova, I. (2026) “Beyond fixed thresholds: optimizing summaries of wearable device data via piecewise linearization of quantile functions.” *Statistics in Medicine*, Vol. 45, No. 15-17, e70646. [Link](#). R package [OptiThresholdR](#).
4. Yeon, K., Gao, F., Kim, R., Lee, D., **Park, J.**, Brown, D.A., and Park, C. (2026) “A Bayesian Approach to Semi-Supervised Domain Adaptation in Streaming Data.” *Journal of Statistical Computation and Simulation*, 96(9), 2237–2262. [Link](#).
3. **Park, J.**, Ahn, J., and Park, C. (2023), “Kernel Sufficient Dimension Reduction and Variable Selection for Compositional Data via Amalgamation.” *International Conference on Machine Learning (ICML)*, pp. 27034–27047, PMLR. [Link](#).
2. Kang, I., Choi, H., Yoon, Y.-J., **Park, J.**, Kwon, S.-S., and Park, C. (2023), “Fréchet Distance-Based Cluster Analysis for Multi-Dimensional Functional Data.” *Statistics and Computing*, 33(4), 75. [Link](#).
1. **Park, J.**, Yoon, C., Park, C., and Ahn, J. (2022), “Kernel Methods for Radial Transformed Compositional Data with Many Zeros.” *International Conference on Machine Learning (ICML)*, pp. 17458–17472, PMLR. [Link](#)

In preparation:

10. **Park, J.**, Szeto, N., Qian, L., Yang, T., Kok, N., and Gaynanova, I. “Multiple Imputation of Missing Predictors and Inference in Fréchet Regression of Distributional Responses.” Planned submission: Oct. 2026.
11. **Park, J.** “Stochastic kernel dimension reduction and divide-and-conquer regression.” Planned submission: Sep. 2026.
12. **Park, J.**, Yeon, H. “Hilbert manifold regression and inference.” Planned submission: Dec. 2026.

TALKS

Invited talks:

- (Scheduled) **ENAR 2027 Spring Meeting** 2027
- **International Conference on Computational Statistics (CompStat)** 2026

- **IMS New Researchers Conference** 2026
- **1st IMS International New Researchers Conference** 2025
- **KAIST, Graduate School of Data Science** 2025

Contributed talks/posters:

- **Mobile Technologies Research Innovation Collaborative (MeTRIC) Symposium** (poster) 2026
 - Beyond fixed thresholds: optimizing summaries of wearable device data
- **IMS International Conference on Statistics and Data Science (ICSDS)** 2025
 - Fast Fréchet regression of multivariate distributions with application to wearable device data
- **Joint Statistical Meetings** 2025
 - Beyond fixed thresholds: optimizing summaries of wearable device data
- **Joint Statistical Meetings** 2024
 - Interpretable dimension reduction for compositional data
- **Winter Conference, the Korean Statistical Society** 2023
 - Interpretable composition-to-composition dimension reduction via conditional covariance operator
- **International Conference on Machine Learning (ICML)** (poster) 2023
 - Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **Summer Conference, the Korean Statistical Society** 2023
 - Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **International Conference on Machine Learning (ICML)** (spotlight talk and poster) 2022
 - Kernel methods for radial transformed compositional data with many zeros
- **2022 Summer Conference, the Korean Statistical Society** 2022
 - Kernel methods for radial transformed compositional data with many zeros (**presentation award**)

ACADEMIC SERVICE

- **Journal Refereeing**
 - Journal of Royal Statistical Society, Series B (1); Journal of American Statistical Association (1); Annals of Applied Statistics (2); Biometrics (3); WIREs Computational Statistics (1)

TEACHING

- **Guest lectures**
 - University of Michigan*
 - (Scheduled twice) Health Applications of Multivariate Analysis (BIOS 690) Fall 2026
 - KAIST*
 - Matrix Group Theory (MAS 435); Topic: connectedness of Lie groups (in English) Spring 2019
 - Modern Algebra II (MAS 312); throughout the semester Fall 2018
- **Teaching Assistant at KAIST**
 - Statistics*
 - Probability and Statistics for Data Science (AI Microdegree Program) Fall 2023
 - Statistical Data Science Practice (DS 516) Spring 2023
 - Probability and Statistics (MAS 250) Fall 2021
 - Mathematical Statistics (MAS 355) Fall 2019
 - Foundational courses (led weekly recitation sections)*
 - Calculus I (MAS 101) Spring 2020
 - Calculus II (MAS 102); *outstanding TA award* in Fall 2020 Fall 2018, Fall 2020

– Introduction to Linear Algebra (MAS 109)

Spring 2018, Spring 2019, Fall 2019, Spring 2021

Mathematics

– Modern Algebra I (MAS 311)

Spring 2021

– Modern Algebra II (MAS 312)

Fall 2018, Fall 2020

– Matrix Group Theory (MAS 435)

Spring 2019

SOFTWARE

- [NonparaTransport](#): an R package for efficient multivariate distributional data analysis
Authors: **Park, J.**, Gaynanova, I.
- [OptiThresholdR](#): an R package for determining optimal thresholds for wearable device measurements
Authors: **Park, J.**, Kok, N., Sousa, F., Gaynanova, I.

LANGUAGES

- English (fluent), Korean (native)

HOBBIES

- Singing, coffee, walking my dog Latte